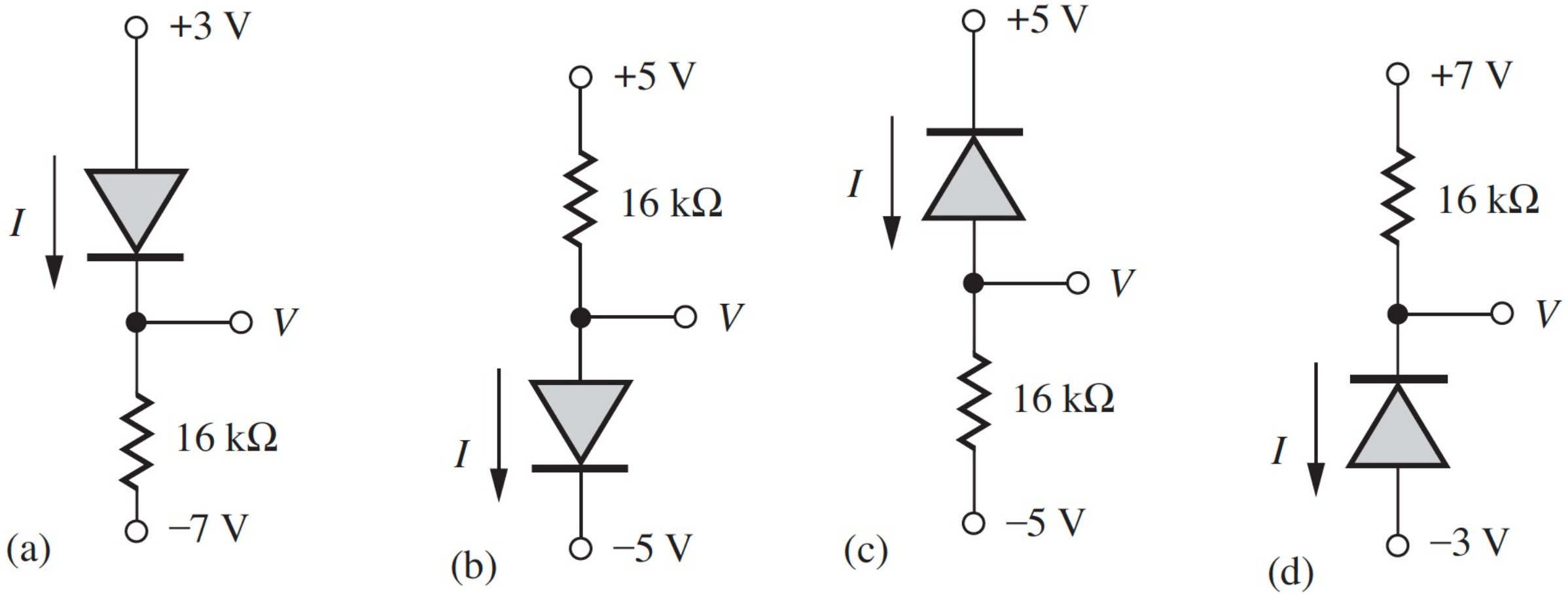
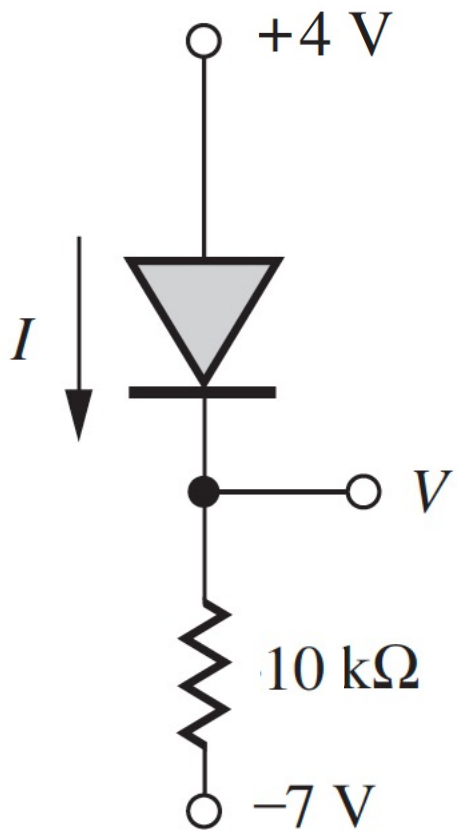
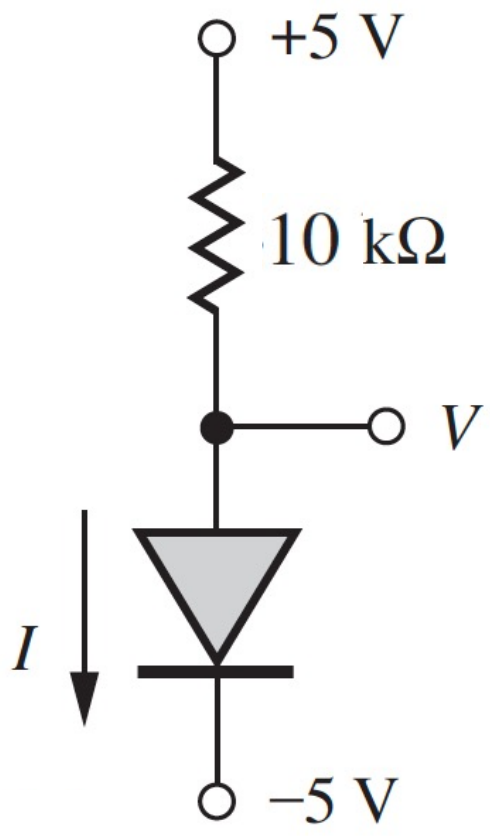
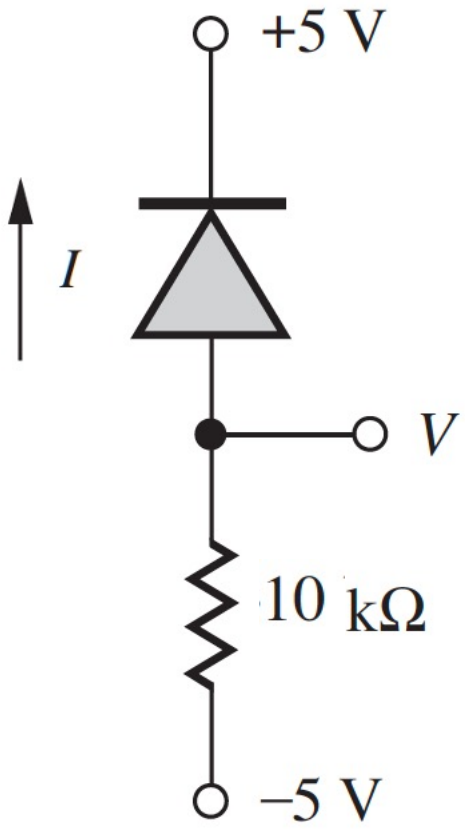
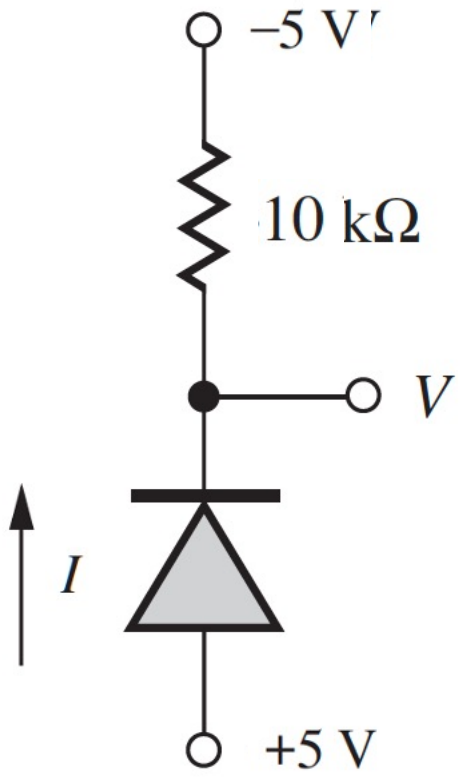

+3 V
I
V
16 kΩ
−7 V
(a)
+5 V
16 kΩ
V
I
−5 V
(b)
+5 V
I
V
16 kΩ
−5 V
(c)
+7 V
16 kΩ
V
I
−3 V
(d)


+4 V
I
V
10 kΩ
−7 V

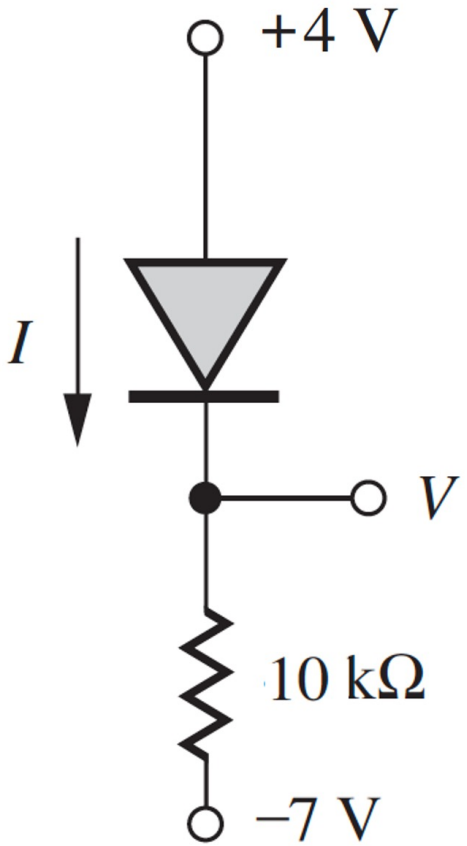

+5 V
10 kΩ
V
I
−5 V


+5 V
I
V
10 kΩ
−5 V

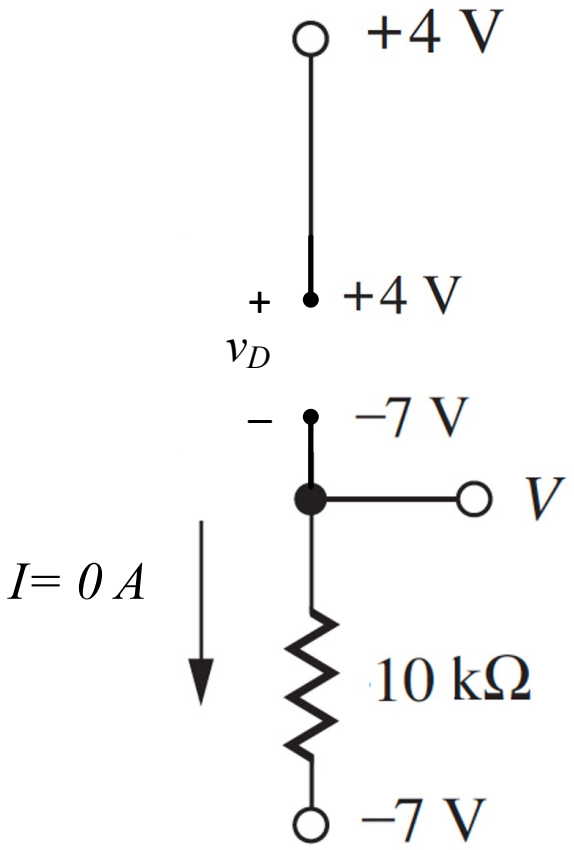

−5 V
10 kΩ
V
I
+5 V

Determinare lo stato del diodo e calcolare I e V.

$V_{on} = 1\ V$



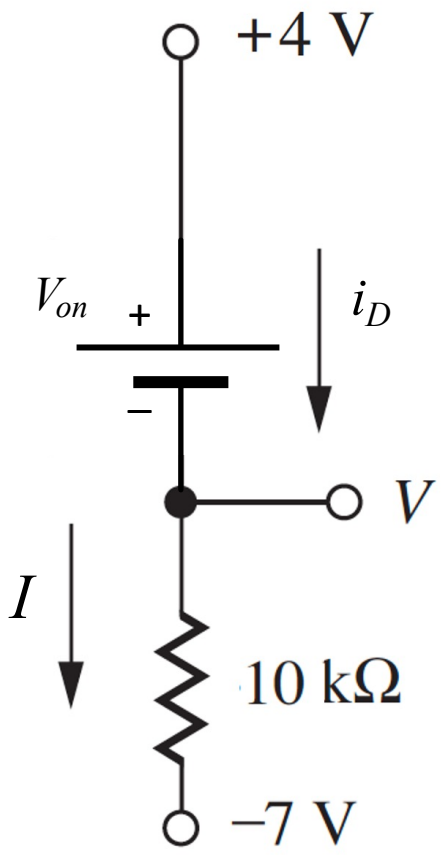
HP Diodo «OFF»



$v_D = 4V - (-7V)$
$= 11V > V_{on}$

→ **Diodo «ON»**

HP Diodo «ON»



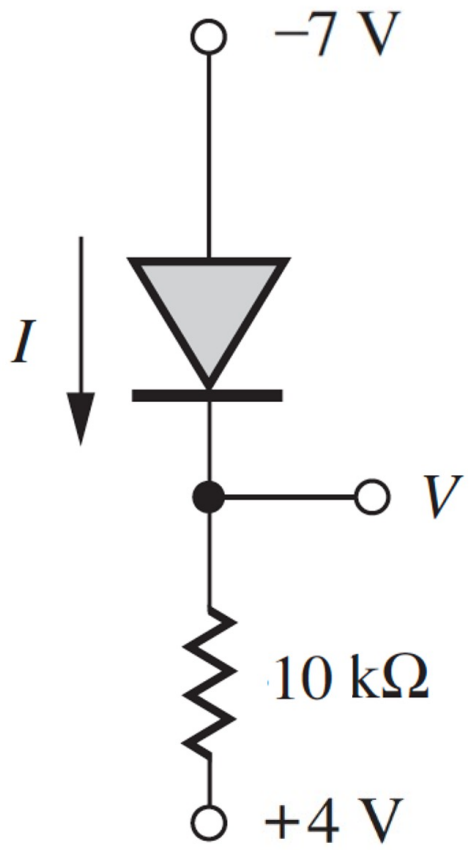
$V = 4V - V_{on} = 3\ V$

$$I = \frac{V - (-7V)}{10\ k\Omega} = \frac{10V}{10\ k\Omega} = 1mA$$

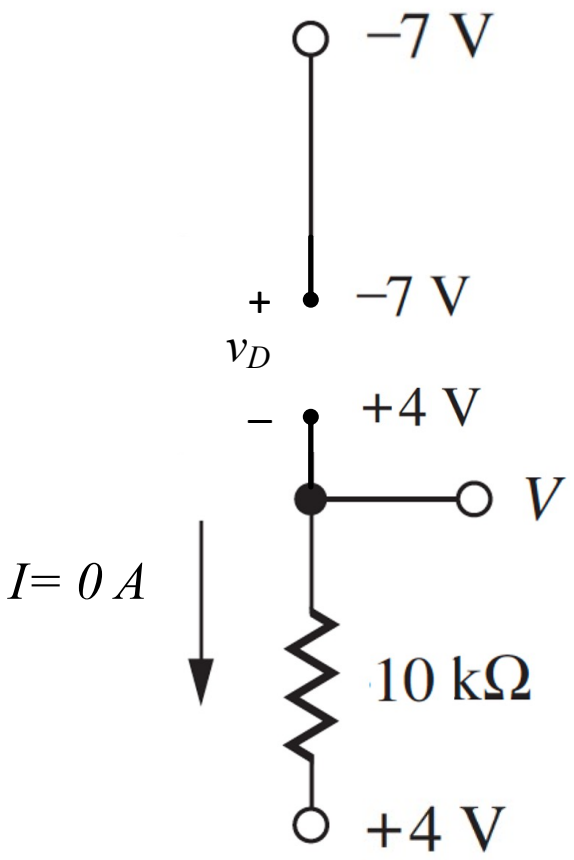
$i_D = I > 0$

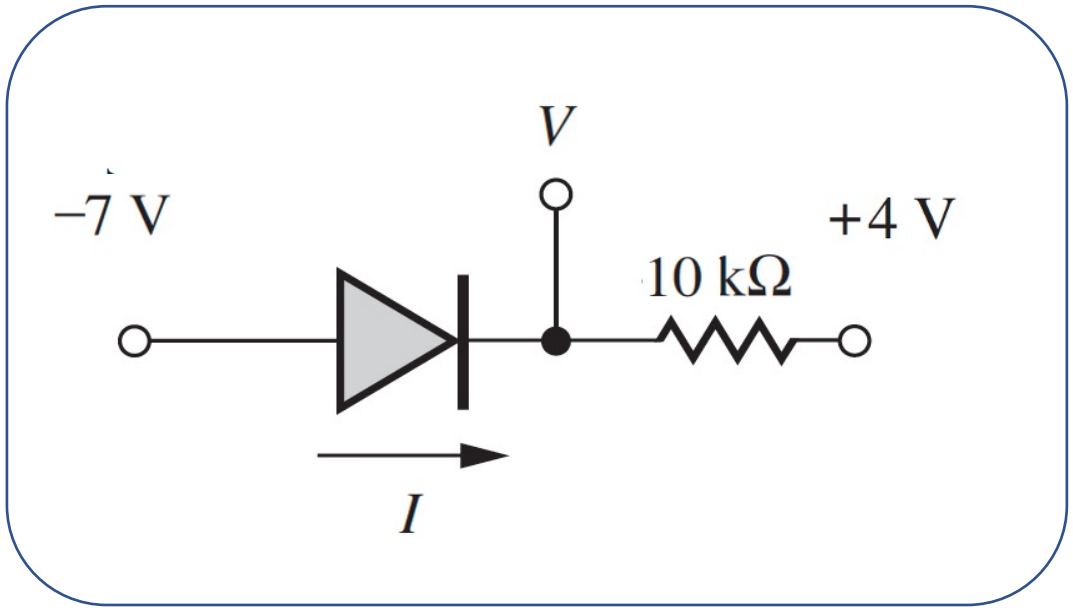
→ **OK Diodo «ON»**

Determinare lo stato del diodo e calcolare I e V.



HP Diodo «OFF»





$$v_D = -7V - 4V$$
$$= -11V < V_{on}$$

→ **OK Diodo «OFF»**

$$V = 4\ V$$

$$I = 0\ mA$$